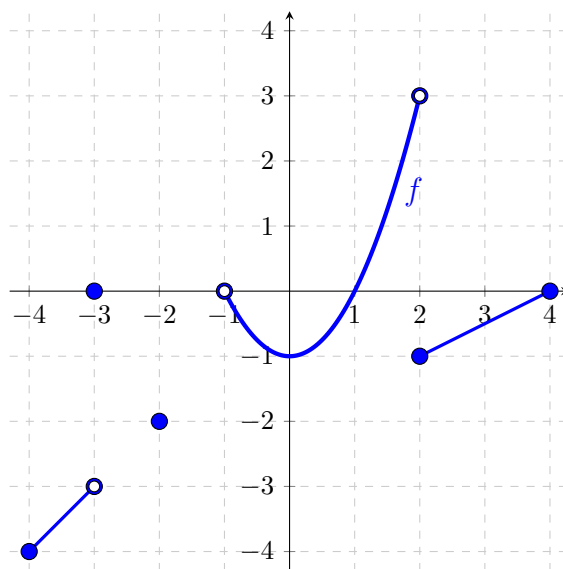


Problem 6

Problem 1 The entire graph of $f(x)$ is given below.



- (a) Find the domain and range of f

Explanation. Domain: $[-4, -3] \cup \{-2\} \cup (-1, 4]$ Range: $[-4, -3] \cup \{-2\} \cup [-1, 3]$

- (b) Find the values of $f(-3)$, $f(-2)$, $f(-1)$, $f(2)$

Explanation. $f(-3) = 0$, $f(-2) = -2$, $f(-1)$ does not exist, $f(2) = -1$

- (c) Find the intervals on which $f(x)$ is positive. Find the intervals on which $f(x)$ is negative.

Explanation. $f(x)$ is positive on $(1, 2)$. $f(x)$ is negative on $[-4, -3)$, $\{-2\}$, $(-1, 1)$, $[2, 4)$

- (d) Find the intervals on which f is increasing. Find the intervals on which f is decreasing.

Explanation. f is increasing on $(-4, -3)$, $(0, 2)$, $(2, 4)$. f is decreasing on $(-1, 0)$. It would also be OK to say that f is increasing on $[-4, -3]$, $[0, 2)$, $[2, 4]$ and f is decreasing on $(-1, 0]$. Typically we will not be including endpoints of intervals when we talk about increasing/decreasing.

Problem 6

(e) *True or False:* $f(1.5) < f(2)$

Explanation. False, $f(2) < f(1.5)$